

q800004

R-ORD

NOT_EQUAL

truncation

questions from

[illegible]

The question

What are the first twenty numbers of the sequence?

the hint the set supplies: the first refers to the smallest

NOT_EQUAL states that these two statements disagree on this data under this rule. It does not state which of them is wrong.

The statements

◀ gold

```
SELECT n FROM sequence ORDER BY n ASC LIMIT 20
```

- ascending, nulls first

```
sha256:3605be48b366c176e0d3bcca06588e5700e13cc25363b4c0
b2ad0e761514367a
```

► **second**

```
SELECT n FROM sequence ORDER BY n ASC LIMIT 8
```

n ascending, nulls first

```
sha256:9312d649b4f63706b2d174037bdb5342ba2a3ab8c2c311a9
d96052bc481af32b
```

The marked tokens are where the two texts differ. Two statements that differ everywhere can return the same rows, and two that differ in one token can return other rows; the verdict above is read off the results.

The rows they differ in

in gold, not in the prediction, 12 rows

from counterexample.json, 12 rows, up to 25 shown per side

side	n INTEGER
◀ gold	10
◀ gold	11
◀ gold	12
◀ gold	13
◀ gold	14
◀ gold	15
◀ gold	16
◀ gold	17
◀ gold	18
◀ gold	19
◀ gold	20
◀ gold	9

in the prediction, not in gold, 0 rows

from counterexample.json, 0 rows, up to 25 shown per side

side	n INTEGER
no rows	

What the benchmark would have said

BIRD's own check: 0

set(second_rows) == set(gold_rows), REAL cells as Python float as sqlite3 returns them, so Python equality holds 1 == 1.0 == True as the benchmark's own scorer does

```
https://github.com/bird-
bench/mini_dev/blob/main/evaluation/evaluation_ex.py
```

```
gold_rows          20
second_rows        8
gold_distinct_rows 20
second_distinct_rows 8
```

the test-suite check: 0

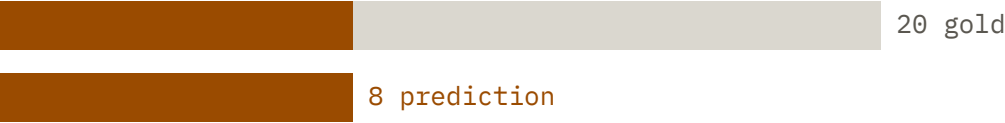
result_eq: equal row counts and equal column counts, each row unordered as a quick rejection, then the two equal as a list when the gold text holds ORDER BY and as a multiset otherwise, under some permutation of the columns; DISTINCT is not stripped and re-executed, and the cells are SQLite's as sqlite3 returns them

```
ruiqi-zhong/test-suite-sql-eval, exec_eval.py,
result_eq, at commit 48cb78ec:
https://github.com/ruiqi-zhong/test-suite-sql-
eval/blob/48cb78ecf7f610620206283846c76751b18a1326/exec
_eval.py

order_matters      true
gold_rows          20
second_rows        8
gold_columns       1
second_columns     1
```

the class: truncation

The class states what makes these two results unequal under this rule, read off the two results and nothing else. It does not state which of the two statements is wrong.



The gold returned 20 rows and the prediction 8, which are the first 8 rows of the other one in the same order.

```
gold_types          ["INTEGER"]
second_types        ["INTEGER"]
multiset_equal      false
set_equal           false
order_equal         false
shorter_result_is_a_prefix true
```

The results

▼ gold, 20 rows

from counterexample.json, 20 rows

n
INTEGER
1
2
3
4
5
6
7
8

n INTEGER
9
10
11
12
13
14
15
16
17
18
19
20

▼ second, 8 rows

from counterexample.json, 8 rows

n INTEGER
1
2
3
4
5
6
7
8

The probes

A smell is a mechanical reason to read this gold statement again. It is a heuristic: it does not state that the statement is wrong, and a maintainer decides.

ordering-over-numeric-text

not applicable

this statement orders by a text column holding only numbers, and ordering it as a number gives a different answer, so the gold may be sorting 9.5 above 10

no ORDER BY key resolves to a text column

▼ what it measured

```
{
  "heuristic": true,
  "reason": "no ORDER BY key resolves to a text column",
  "keys": [
    {
      "key": "n",
      "column": "sequence.n",
      "declared_type": "INTEGER",
      "not_applicable": "the column is not declared as text"
    }
  ]
}
```

this statement cuts its result at a LIMIT that does not decide which rows come back, so a different but equally correct statement can return other rows and score zero

▼ what it measured

```
{
  "heuristic": true,
  "cut": 20,
  "offset": 0,
  "distinct_kept": false,
  "unbounded_sql": "SELECT n, n AS attestql_ordering_key_0 FROM sequence ORDER BY n ASC",
  "unbounded_rows": 40,
  "projected_columns": [
    "n"
  ],
  "ordering_key_columns": [
    "attestql_ordering_key_0"
  ],
  "ordering_keys": [
    {
      "key": "n",
      "direction": "asc",
      "nulls": "first",
      "nulls_first_in_effect": true,
      "returned_rows_null_in_this_key": 0,
      "fires": false
    }
  ],
  "case": null
}
```

rerun over the same rows in another physical order this statement gives another answer, so its result depends on how the rows are stored and not only on the data

▼ what it measured

```
{
  "heuristic": true,
  "rule": "R-ORD",
  "baseline_result_hash":
"sha256:3605be48b366c176e0d3bcca06588e5700e13cc25363b4c0b2ad0e761514367a",
  "baseline_result": {
    "columns": [
      {
        "name": "n",
        "declared_type": "INTEGER"
      }
    ],
    "row_count": 20,
    "truncated": false,
    "rows_shown": 10,
    "rows": [
      [
        {
          "type": "int",
          "value": 1
        }
      ],
      [
        {
          "type": "int",
          "value": 2
        }
      ],
      [
        {
          "type": "int",
          "value": 3
        }
      ],
      [
        {
          "type": "int",
          "value": 4
        }
      ],
      [
        {
          "type": "int",
          "value": 5
        }
      ]
    ]
  }
}
```

```
],
[
  {
    "type": "int",
    "value": 6
  }
],
[
  {
    "type": "int",
    "value": 7
  }
],
[
  {
    "type": "int",
    "value": 8
  }
],
[
  {
    "type": "int",
    "value": 9
  }
],
[
  {
    "type": "int",
    "value": 10
  }
]
],
"result_hash": "sha256:3605be48b366c176e0d3bcca06588e5700e13cc25363b4c0b2ad0e761514367a"
},
"planner_statistics": {},
"shuffle": {
  "seed": "1",
  "row_limit": 300000,
  "tables": [
    "sequence"
  ],
  "tables_not_shuffled": [],
  "tables_skipped_for_size": {},
  "tables_not_reached_by_a_copy": {}
},
"shuffled_copies": {
  "run": true,
  "verdict": "equal",
  "differs": false,
  "result_hash": "sha256:3605be48b366c176e0d3bcca06588e5700e13cc25363b4c0b2ad0e761514367a",
  "result": {
    "columns": [
      {
        "name": "n",
```



```
"declared_type": "INTEGER"
}
],
"row_count": 20,
"truncated": false,
"rows_shown": 10,
"rows": [
  [
    {
      "type": "int",
      "value": 1
    }
  ],
  [
    {
      "type": "int",
      "value": 2
    }
  ],
  [
    {
      "type": "int",
      "value": 3
    }
  ],
  [
    {
      "type": "int",
      "value": 4
    }
  ],
  [
    {
      "type": "int",
      "value": 5
    }
  ],
  [
    {
      "type": "int",
      "value": 6
    }
  ],
  [
    {
      "type": "int",
      "value": 7
    }
  ],
  [
    {
      "type": "int",
      "value": 8
    }
  ]
]
```

```
],
[
  {
    "type": "int",
    "value": 9
  }
],
[
  {
    "type": "int",
    "value": 10
  }
]
],
"result_hash": "sha256:3605be48b366c176e0d3bcca06588e5700e13cc25363b4c0b2ad0e761514367a"
}
},
"plan_variant": {
  "run": false,
  "reason": "the plan variant was not asked for"
}
}
```

The evidence records

gold: evidence-gold.json

```
SELECT n FROM sequence ORDER BY n ASC LIMIT 20
```

```
statement read from /private/tmp/claude-501/-Users-
hoangle-orca-workspaces-attestdql-web-
ui/a7918775-abd6-4451-9c1e-
f770717fa188/scratchpad/stress/questi
ons.json
```

```
digest      sha256:fe74363312c646fcbc137792176785
            89d60ff6c240fd23e80535474e3d482003
```

[illegible]

```
result hash
```

```
sha256:3605be48b366c176e0d3bccca06588e5700e13cc25363b4c0
```

b2ad0e761514367a recomputed from this JSON: match

record_hash

```
sha256:5242e888fecf7abc576356d08f7d10f9972be99870b32f0f
```

dfbbcd8ecc24ac49 recomputed from this JSON: match

▼ the result this record holds, 20 rows

from evidence-gold.json, 20 rows

n
INTEGER
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19

n
INTEGER
20

▼ what ran, and where

run	audit-ef32e668-f58f-44e0-b835-dcd5fa4fbc28
executed at	2026-09-08T02:36:48.700391+00:00
data as of	2026-09-08T02:36:48.656491+00:00
backend at checkout	SQLite 3.53.4 file=/private/tmp/claude-501/- Users-hoangle-orca-workspaces- attestql-web-ui/a7918775-abd6-4451- 9c1e- f770717fa188/scratchpad/stress/fixt ure.sqlite size=2252800
backend that answered	SQLite 3.53.4 file=/private/tmp/claude-501/- Users-hoangle-orca-workspaces- attestql-web-ui/a7918775-abd6-4451- 9c1e- f770717fa188/scratchpad/stress/fixt ure.sqlite size=2252800
database role	file
replay rule	R-ORD
question set version	sha256:fe74363312c646fcbc1377921767 8589d60ff6c240fd23e80535474e3d48200 3
validator	audit:sqlglot-sqlite-parse
checks run	parses_as_exactly_one_statement, the_one_statement_is_a_select, no_placeholder_without_a_bound_para meter
statement timeout	2000 ms
rows	20 rows

▼ the session it ran under

engine	sqlite
time_zone	not stated by this engine
date_style	not stated by this engine
interval_style	not stated by this engine
extra_float_digits	not stated by this engine
database_collation	not stated by this engine
work_mem	not stated by this engine
hash_mem_multiplier	not stated by this engine

recorded beside them

sqlite_version	3.53.4
encoding	UTF-8
reverse_unordered_selects	0
query_only	1
journal_mode	delete
data_version	1

compile_options	ATOMIC_INTRINSICS=1, COMPILER=c1a ng- 21.0.0, DEFAULT_AUTOVACUUM, DEFAULT_CACHE_SIZE=-2000, DEFAULT_FILE_FORMAT=4, DEFAULT_JOURNAL_SIZE_LIMIT=-1, DEFAULT_MMAP_SIZE=0, DEFAULT_PAGE_SIZE=4096, DEFAULT_PCACHE_INITSZ=20, DEFAULT_RECURSIVE_TRIGGERS, DEFAULT_SECTOR_SIZE=4096, DEFAULT_SYNCHRONOUS=2, DEFAULT_WAL_AUTOCHECKPOINT=1000, DEFAULT_WAL_SYNCHRONOUS=2, DEFAULT_WORKER_THREADS=0, DIRECT_OVERFLOW_READ, ENABLE_API_ARMOR, ENABLE_COLUMN_METADATA, ENABLE_DBSTAT_VTAB, ENABLE_FTS3, ENABLE_FTS3_PARENTHESIS, ENABLE_FTS5, ENABLE_GEOPOLY, ENABLE_MATH_FUNCTIONS, ENABLE_MEMORY_MANAGEMENT, ENABLE_PERCENTILE, ENABLE_PREUPDATE_HOOK, ENABLE_RTREE, ENABLE_SESSION, ENABLE_STAT4, ENABLE_UNLOCK_NOTIFY, MALLOC_SOFT_LIMIT=1024, MAX_ATTACHED=10, MAX_COLUMN=2000, MAX_COMPOUND_SELECT=500, MAX_DEFAULT_PAGE_SIZE=8192, MAX_EXPR_DEPTH=1000, MAX_FUNCTION_ARG=1000, MAX_LENGTH=1000000000, MAX_LIKE_PATTERN_LENGTH=50000, MAX_MMAP_SIZE=0x7fff0000, MAX_PAGE_COUNT=0xfffffff, MAX_PAGE_SIZE=65536, MAX_SQL_LENGTH=1000000000, MAX_TRIGGER_DEPTH=1000, MAX_VARIABLE_NUMBER=250000, MAX_VDBE_OP=250000000, MAX_WORKER_THREADS=8, MUTEX_PTHREADS, SYSTEM_MALLOC, TEMP_STORE=1, THREADSAFE=1, USE_URI
collation_list	RTRIM, NOCASE, BINARY
case_sensitive_like	0

▼ the rendering and the data

version	attestql/audit/2
numeric_scale	6
timestamp_format	%Y-%m-%dT%H:%M:%S.%fZ
timezone	UTC
null_rendering	NULL
encoding	utf-8
schema_digest	sha256:e2abf655497c07bbcf5c487e0cd606ddd5e6ead2d1e44d6dc73e17497959d32b
source_file_sha256	
rows in main.sequence	40

second: evidence-second.json

```
SELECT n FROM sequence ORDER BY n ASC LIMIT 8
```

statement read from /private/tmp/claude-501/-Users-hoangle-orca-workspaces-attestql-web-ui/a7918775-abd6-4451-9c1e-f770717fa188/scratchpad/stress/predictions.json

digest sha256:8c0123478e1a1041b978bb306240e8fd8ef0c5eba51615f537595543e1250ff3

origin the run was told none

result_hash

sha256:9312d649b4f63706b2d174037bdb5342ba2a3ab8c2c311a9d96052bc481af32b recomputed from this JSON: match

record_hash

sha256:eacbb1a2b6e79bffc8784987da7e6d926c2e3da383c785416a4c12bbee128b69 recomputed from this JSON: match

▼ the result this record holds, 8 rows

from evidence-second.json, 8 rows

n
INTEGER
1
2
3
4
5
6
7
8

▼ what ran, and where

run audit-ef32e668-f58f-44e0-b835-dcd5fa4fbc28

executed at 2026-09-08T02:36:48.700454+00:00

data as of 2026-09-08T02:36:48.656491+00:00

backend at checkout SQLite 3.53.4 | file=/private/tmp/claude-501/-Users-hoangle-orca-workspaces-attestql-web-ui/a7918775-abd6-4451-9c1e-f770717fa188/scratchpad/stress/fixture.sqlite | size=2252800

backend that answered SQLite 3.53.4 | file=/private/tmp/claude-501/-Users-hoangle-orca-workspaces-attestql-web-ui/a7918775-abd6-4451-9c1e-f770717fa188/scratchpad/stress/fixture.sqlite | size=2252800

database role	file
replay rule	R-ORD
question set version	sha256:fe74363312c646fcbc13779217678589d60ff6c240fd23e80535474e3d482003
validator	audit:sqlglot-sqlite-parse
checks run	parses_as_exactly_one_statement, the_one_statement_is_a_select, no_placeholder_without_a_bound_parameter
statement timeout	2000 ms
rows	8 rows

▼ the session it ran under

engine	sqlite
time_zone	not stated by this engine
date_style	not stated by this engine
interval_style	not stated by this engine
extra_float_digits	not stated by this engine
database_collation	not stated by this engine
work_mem	not stated by this engine
hash_mem_multiplier	not stated by this engine

recorded beside them

sqlite_version	3.53.4
encoding	UTF-8
reverse_unordered_selects	0
query_only	1
journal_mode	delete
data_version	1
compile_options	ATOMIC_INTRINSICS=1,COMPILER=clang-21.0.0,DEFAULT_AUTOVACUUM,DEFAULT_CACHE_SIZE=-2000,DEFAULT_FILE_FORMAT=4,DEFAULT_JOURNAL_SIZE_LIMIT=-1,DEFAULT_MMAP_SIZE=0,DEFAULT_PAGE_SIZE=4096,DEFAULT_PCACHE_INITSZ=20,DEFAULT_RECURSIVE_TRIGGERS,DEFAULT_SECTOR_SIZE=4096,DEFAULT_SYNCHRONOUS=2,DEFAULT_WAL_AUTOCHECKPOINT=1000,DEFAULT_WAL_SYNCHRONOUS=2,DEFAULT_WORKER_THREADS=0,DIRECT_OVERFLOW_READ,ENABLE_API_ARMOR,ENABLE_COLUMN_METADATA,ENABLE_DBSTAT_VTAB,ENABLE_FTS3,ENABLE_FTS3_PARENTHESIS,ENABLE_FTS5,ENABLE_GEOPOLY,ENABLE_MATH_FUNCTIONS,ENABLE_MEMORY_MANAGEMENT,ENABLE_PERCENTILE,ENABLE_PREUPDATE_HOOK,ENABLE_RTREE,ENABLE_SESSION,ENABLE_STAT4,ENABLE_UNLOCK_NOTIFY,MALLOC_SOFT_LIMIT=1024,MAX_ATTACHED=10,MAX_COLUMN=2000,MAX_COMPOUND_SELECT=500,MAX_DEFAULT_PAGE_SIZE=8192,MAX_EXPR_DEPTH=1000,MAX_FUNCTION_ARG=1000,MAX_LENGTH=1000000000,MAX_LIKE_P

	ATTERN_LENGTH=50000,MAX_MMAP_SIZE=0x7fff0000,MAX_PAGE_COUNT=0xffffffe,MAX_PAGE_SIZE=65536,MAX_SQL_LENGTH=1000000000,MAX_TRIGGER_DEPTH=1000,MAX_VARIABLE_NUMBER=250000,MAX_VDBE_OP=250000000,MAX_WORKER_THREADS=8,MUTEX_PTHREADS,SYSTEM_MALLOC,TEMP_STORE=1,THREADS SAFE=1,USE_URI
collation_list	RTRIM,NOCASE,BINARY
case_sensitive_like	0

▼ the rendering and the data

version	attestql/audit/2
numeric_scale	6
timestamp_format	%Y-%m-%dT%H:%M:%S.%fZ
timezone	UTC
null_rendering	NULL
encoding	utf-8
schema digest	sha256:e2abf655497c07bbcf5c487e0cd606ddd5e6ead2d1e44d6dc73e17497959d32b
source file sha256	
rows in main.sequence	40

Running these again

gold

re-run this statement read-only against SQLite 3.53.4 |
file=/private/tmp/claude-501/-Users-hoangle-orca-workspaces-attestql-web-ui/a7918775-abd6-4451-9c1e-f770717fa188/scratchpad/stress/fixture.sqlite | size=2252800 under the session settings and over the data this record's fixture digest names, and compare the two results under R-ORD

second

re-run this statement read-only against SQLite 3.53.4 |
file=/private/tmp/claude-501/-Users-hoangle-orca-workspaces-attestql-web-ui/a7918775-abd6-4451-9c1e-f770717fa188/scratchpad/stress/fixture.sqlite | size=2252800 under the session settings and over the data this record's fixture digest names, and compare the two results under R-ORD

This question's run

run audit-ef32e668-f58f-44e0-b835-dcd5fa4fbc28

[illegible]

the run this question belongs to <../index.html>

The JSON this page was rendered from

```
counterexample.json <counterexample.json>
attestql/audit/counterexample/2
```

evidence-gold.json <evidence-gold.json> the gold's evidence record

`evidence-second.json` <`evidence-second.json`> the prediction's evidence record

```
smells.json <smells.json> attestql/audit/smells/2
```